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24 December 1980
Superseding
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26 June 1962

MILITARY SPECIFICATION

CONTROL, ANTENNA C-1482/M

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of antenna control designated as Control, Antenna C-1482/M.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-T-945	Test Equipment, For Use With Electronic Equipment: General Specification
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

Beneficial comments (recommendations, additions, deletions) any any pertinent data which may be of use in improving this document should be addressed to: Commander, USA Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1430

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STANDARDS

Military

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-202	Test Methods for Electronics and Electrical Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects for
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

DRAWINGS

Army

7601903	Synchro Dial and Index Marking
7616154	Synchro Units Zero Adjust
7676906	Synchro Control Transmitter Type 23CX4
8008199	Control, Antenna C-1482/M

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Preproduction sample. When a preproduction sample is required (see 6.2), it shall be manufactured using the same methods proposed for the manufacture of subsequent production lots.

3.2 Material. All material used in the construction of each antenna control shall be in accordance with Drawing 8008199 and applicable drawings and specifications referenced thereon.

3.3 Design and construction. The design and construction of each antenna control shall be in accordance with Drawing 8008199 and applicable drawings and specifications referenced thereon.

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3.4 Electrical.

3.4.1 Ground potential. All external metallic parts shall be at ground potential at all times.

3.4.2 Circuitry resistance. The circuitry resistance of each antenna control shall be in accordance with Drawing 8008253 covering a detail of Drawing 8008199.

3.4.3 Dielectric strength. There shall be no sparkover or decrease in dielectric strength when the insulated areas of the antenna control are subjected to an alternating current (ac) root mean square (rms) potential of 500 volts, plus or minus 25 volts, at a frequency of 60 cycles per second (cps), plus or minus 3 cps, for a period of not less than 3 nor more than 5 seconds.

3.4.4 Insulation resistance. Uness otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits shall have a minimum resistance of 100 megohms.

3.5 Performance characteristics.

3.5.1 Relay circuit operation.

3.5.1.1 Relay input voltage. Relays, K1, K2, and K3, shall be capable of operating as specified herein when supplied with input voltage as specified in table I.

3.5.1.2 Relay power consumption. The power consumption of relays K1, K2, and K3, shall be such that the total energizing current shall be not greater than that specified in table I.

Table I - Input Voltage and Power Consumption

Input Voltage	Input Connection	Current
Negative 28 volts, direct current (dc), plus or minus 1 volt	Terminal P1, Pin 14 (negative) and Pin 13 (ground)	220 milli-ampere (ma)
400 cps, ac, single phase voltage, adjustable to 100 volts	Terminal P1, and 12 and 15	10 ma
120 volts, plus or minus 1.2 volts, ac, 400 cps single phase 1.0 ampere	Terminals R1 and R2 of synchros B1 and B2, and external synchro control transmitter	

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3.5.1.3 Relay circuit continuity. Relay circuit continuity (zero resistance) shall exist between the terminal P1 pins and relay terminal pins specified in table II, when relays K1, K2, and K3 are energized. Open circuit (infinite resistance) shall exist between the pins specified when the relay energizing voltage is removed.

Table II

Terminal P1 pin	Relay Terminal	Relay
2	9	
1	10	K1
3	11	
5	12	
6	9	
7	10	K2
9	11	
11	12	
10	12	K3
15	11	

3.5.2 Range and synchro circuits.

3.5.2.1 Input voltage. The range circuit, and synchros B1 and B2, shall be capable of operating as specified herein when supplied with input voltages as specified in table I.

3.5.2.2 Power consumption. Power consumption of the range and synchro circuits shall be such that the total current shall be not greater than that specified in table I.

3.5.2.3 Range variable resistor. The range variable resistor, R1, shall be capable of adjustment to provide a voltage at terminal P1, pins 10 and 12, equal to 1 volt for each 100 yards of dial reading, plus or minus 3 percent.

3.5.2.4 Synchro zero adjust. The antenn control synchros, B1 and B2, shall be capable of adjustment to provide electrical zero (less than 0.2 volt).

3.5.2.5 Elevation and azimuth synchro operation. Synchros B1 and B2 shall be capable of synchronization with an external synchro control transmitter.

3.6 Environmental and service conditions.

3.6.1 Temperature.

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3.6.1.1 Operating. The antenna control shall operate satisfactorily within ambient temperatures ranging from minus 40 degrees to plus 51.7 degrees centigrade (c) (minus 40 degrees to plus 125 degrees Fahrenheit (F).

3.6.1.2 Nonoperating. The antenna control shall be capable of being stored and transported within damage or impairment of performance in ambient temperatures ranging from minus 45 degrees to plus 60 degrees C (minus 49 degrees to plus 140 degrees F).

3.6.2 Vibration. The antenna control shall not be damaged nor shall its performance be impaired when subjected to vibrating frequencies in the range of 10 to 55 cps, total excursion 0.0156 inch plus or minus 10 percent.

3.6.3 Shock. The antenna control shall be capable of withstanding without damage, the shocks that occur in normal operating and nonoperating conditions.

3.6.4 Relative humidity. The antenna control shall not be damaged nor shall its performance be impaired when subjected to relative humidities up to and including 90 percent.

3.6.5 Fungus, resistance. The antenna control shall be treated in accordance with MIL-T-152.

3.6.6 Salt-laden air. The antenna control shall not be damaged nor shall its performance be impaired by salt-laden air as encountered in coastal service or in ocean transportation.

3.6.7 Altitude. The antenna control shall operate satisfactorily at any altitude from sea level to 10,000 feet above sea level. The unit, non-operating, shall be capable of being transported by air, without damage, at altitudes up to and including 30,000 feet.

3.7 Details of components. Components of each antenna control shall be in accordance with Drawing 8008199 and applicable drawings and specifications referenced thereon.

3.8 Dimensions. Dimensions and tolerances of each antenna control shall be in accordance with Drawing 8008199 and applicable drawings and specifications referenced thereon.

3.9 Color and finish. Color and finish of each antenna control shall be in accordance with Drawing 8008199 and applicable drawings and specifications referenced thereon.

3.10 Product marking. Product marking shall be in accordance with Drawing 8008199, applicable drawings referenced thereon, and MIL-STD-130.

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3.11 Workmanship. All components of the antenna control shall be fitted properly and firmly into their respective positions. All mechanical joints and electrical connections shall be mechanically secure and electrically sound. All controls shall operate freely without slipping, sticking or binding. Electrical connections shall be flexible between shock mounted components and their foundations. Surfaces of glazed ceramic components shall be smooth, uniform, and free from cracks. There shall be no evidence of corrosion, burrs, roughness, sharp edges or other defects which could adversely affect its use or sutiability for the purpose intended.

4. QUALITY ASSURANCE

4.1 Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. Unless otherwise specified (see 6.2), the preproduction sample shall be subjected to all examinations and tests specified herein. The Government will perform the examinations and tests at a place specified by the procuring activity.

4.3 Inspection. Examination and tests shall be conducted on 100 percent of production with the exception of environmental.

4.4 Examination.

4.4.1 Visual and tactile. The antenna controls shall be given a visual and tactile examination for defects to determine compliance with 3.2, 3.3, 3.7, 3.8, 3.9, 3.10 and 3.11. MIL-STD-252 shall be used where applicable.

4.4.2 Inspection of preparation for delivery. Preservation, packaging and packing shall be inspected for conformance to the requirements of Section 5.

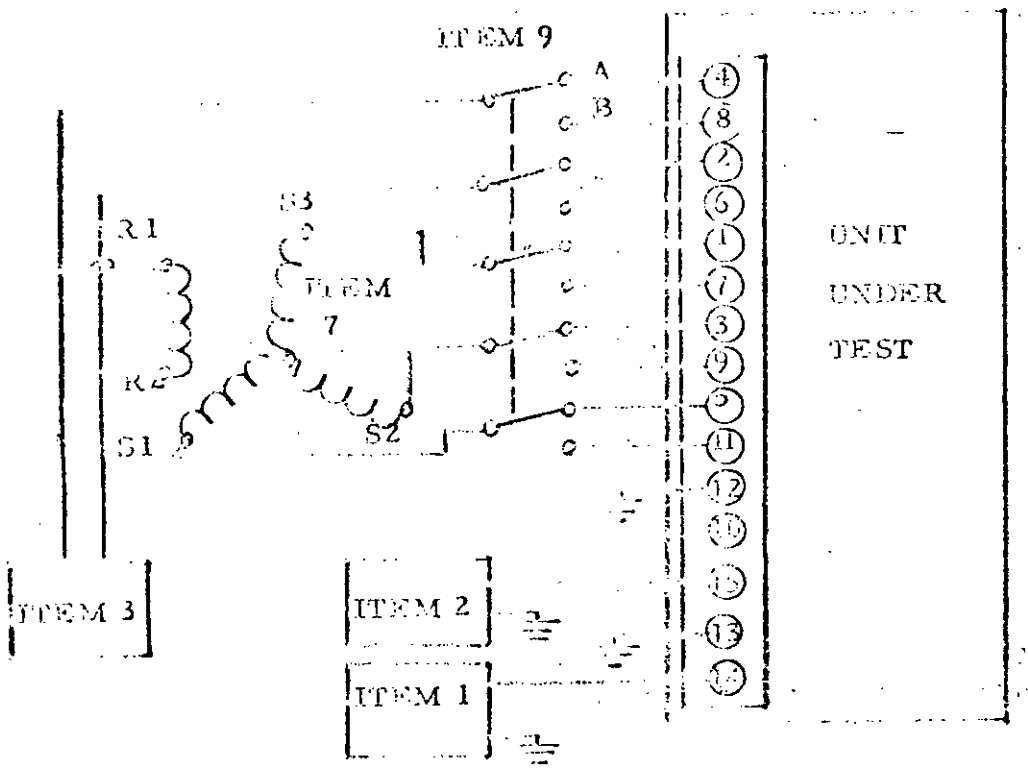
4.5 Test equipment. The test equipment shall meet the minimum requirements of the equipment listed below and shall be used as indicated in performing the tests. In the test setup and procedure the equipment is referred to by item number. The equipment may be mounted separately or in any combination (such as consoles, panels, and multi-purpose instruments).

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ITEM	DESCRIPTION
1	Power supply, dc, negative 28 volts, plus or minus 1 volt, 220 ma.
2	Power supply, ac, 100 volts, 400 cps, single phase, 10 ma.
3	Power supply, ac, 120 volts, 400 cps, single phase, 1.0 ampere.
4	Voltmeter, vacuum tube, capable of measuring an ac voltage in the range of 0 to 150 volts and a dc voltage in the range of 0 to 50 volts.
5	Multimeter, TS-372B/U or equivalent.
6	Milliammeter, capable of measuring ac current in the range of 0 to 1 ampere.
7	Synchro control transmitter, type 23CX4, conforming to Drawing 7676906, or equivalent.
8	Synchroidal and index marking, conforming to Drawing 7601903 or equivalent.
9	Switch; 5 pole 2 position.

4.6 Test setup. The test setup indicated in Figure 1 shall be used as a guide in conducting the performance characteristic tests.

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4.7 Tests. All terminals shall be grounded or connected as specified. Tests specified in 4.7.1 shall be performed prior to application of power. Tests specified in 4.7.2 shall be performed in the sequence indicated. Power consumption and environmental and service condition tests shall be performed on the preproduction sample only.

4.7.1 Electrical.

4.7.1.1 Ground potential. All external metallic parts shall be measured to ground using item 5 to determine compliance with 3.4.1.

4.7.1.2 Circuitry resistance. A circuitry resistance check of each antenna control shall be made using item 5 to determine compliance with 3.4.2.

4.7.1.3 Dielectric strength. Each antenna control shall be tested in accordance with MIL-STD-202, method 301, to determine compliance with 3.4.3.

4.7.1.4 Insulation resistance. Each antenna control shall be given an insulation resistance test in accordance with MIL-STD-202, method 302, condition A, to determine compliance with 3.4.4.

4.7.2 Performance characteristics.

4.7.2.1 Relay circuit operation. The measurements specified in 4.7.2.1.1, 4.7.2.1.2, and 4.7.2.1.3 shall be made with relay input voltage only applied as specified in 3.5.1.1. All other power supplies and test equipment shall be disconnected from the unit under test.

4.7.2.1.1 Relay input voltage. Item 4 shall be used to measure the relay input voltage to determine compliance with 3.5.1.1.

4.7.2.1.2 Relay power consumption. Item 5 shall be used to measure the relay energizing current to determine power consumption as specified in 3.5.1.2.

4.7.2.1.3 Relay circuit continuity. Item 5 shall be used to measure the relay circuit resistance between the terminal P1 pins and relay terminal pins specified in table II. Measurements shall be made with the relay in an energized condition and then in a de-energized condition to determine compliance with 3.5.1.3.

4.7.2.2 Range and synchro circuits.

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4.7.2.2.1 Input voltage. All power supplies and test equipment indicated in figure 1 shall be connected to the unit under test, and the range and synchro circuit input voltages applied as specified in table I. Item 4 shall be used to measure the input voltages to determine compliance with 3.5.2.1.

4.7.2.2.2 Power consumption. Item 6 shall be used to measure the range and synchro circuit current to determine power consumption as specified in 3.5.2.2.

4.7.2.2.3 Range variable resistor. The antenna control range dial shall be set at 10,000 yards. Items 2 shall be adjusted to provide 100 volts at terminal P1 pins 10 and 12 and measured using item 4. The range dial shall then be ajusted to 2000, 4000, 6000, and 8000 yard settings in succession and the output voltage at terminal P1 pins 10 and 12 measured at each setting using item 4 to determine compliance with 3.5.2.3.

4.7.2.2.4 Synchros zero adjust. Item 8 (synchrodial) shall be mounted on item 7 (external synchro control transmitter). The elevation and azimuth synchros, B1 and B2, and item 7 shall be adjusted for electrical zero as specified on Drawing 7616154, with all synchrodials set to zero, to determine compliance with 3.5.2.4.

4.7.2.2.5 Elevation and azimuth synchro operation. Item 9 shall be set to position A. Item 8 shall be rotated throughout the entire dial range and the elevation synchrodial shall be observed to follow in synchroization to determine compliance with 3.5.2.5. Item 9 shall then be set to position B and the above test repeated on the azimuth circuit to determine compliance with 3.5.2.5.

4.7.3 Environmental and service conditions.

4.7.3.1 Temperature. The operating and nonoperating temperature tests shall be performed in accordance with the high and low temperature tests of MIL-E-5272, procedure II, with the exception that the extremes of temperature to which the antenna control shall be subjected shall be as specified in 3.6.1.

4.7.3.2 Vibration. The antenna control shall be tested in accordance with MIL-T-945, vibration test 2, except the unit shall be in nonoperating condition and the total vibration excursion shall be 0.0156 inch, plus or minus 10 percent (see 3.6.2).

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4.7.3.3 Shock. The antenna control, nonoperating, shall be titled up through an angle of 30 degrees or 6 inches whichever is less, using one edge of the unit as an axis. The unit shall be allowed to fall back freely to the horizontal position on a 2-inch-thick hardwood surface. The test shall be repeated using other practical edges of the same horizontal face as axes, for a total of 4 drops. The unit shall be checked for proper operation following the series of shock tests (see 3.6.3).

4.7.3.4 Relative humidity. The antenna control shall be tested in accordance with MIL-E05272, procedure III, except the relative humidity shall be 90 percent as specified in 3.6.4.

4.7.3.5 Fungus resistance. The antenna control shall be tested in accordance with the fungus resistance test of MIL-E-5272, procedure I, except that no evidence of fungus growth shall be present upon conclusion of the test (see 3.6.5).

4.7.3.6 Salt-laden air. The antenna control shall be tested in accordance with the salt spray test of MIL-E-5272, procedure I, to determine compliance with 3.6.6.

4.7.3.7 Altitude. The antenna control shall be tested in accordance with the temperature-altitude test of MIL-E-5272, procedure II. The unit shall be in nonoperating condition for the 30,000 foot test and the ambient temperature shall be within 5 degrees of the extremes specified in 3.6.1 for each condition of 3.6.7.

5. PACKAGING

5.1 Preservation, packing, unitization, and marking. Level A, B or commercial packaging (see 6.2).

5.1.1 Level A or B. Preservation, packing, unitization, and marking shall be in accordance with packaging data sheet APN 8008199 and specifications referenced therein.

5.1.2 Commercial. Preservation, packing, unitization and marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Control, Antenna C-1482/M is used to provide preset coordination information to be used for slewing the missile track radar to a predetermined location of beacon (NIKE Missile System).

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6.2 Ordering data. Procurement document should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1).

c. Number of preproduction samples required.

d. Where the preproduction sample tests shall be performed (see 4.2).

e. Selection of applicable levels of preservation, packaging and packing (see 5.1).

f. Any special marking required (see 5.1).

g. Applicable stock number.

Custodian:
Army-MI

Preparing activity:
Army-MI

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-C-45776A(MI) CONTROL, ANTENNA C-1482/M

NAME OF ORGANIZATION AND ADDRESS

—

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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